

## DATE & TIMESTAMP FUNCTIONS IN MYSQL (DETAILED)

### 1. Sample Table Structure

```
CREATE TABLE orders (  
order_id INT PRIMARY KEY,  
customer_name VARCHAR(50),  
order_date DATE,  
order_timestamp TIMESTAMP,  
delivery_date DATE,  
order_amount DECIMAL(10,2)  
);
```

### 2. Insert Sample Data

```
INSERT INTO orders VALUES  
(1, 'Karthik', '2024-01-15', '2024-01-15 10:30:45', '2024-01-20', 2500.00),  
(2, 'Veena', '2024-02-18', '2024-02-18 18:45:20', '2024-02-22', 3200.50),  
(3, 'Ravi', '2024-03-02', '2024-03-02 09:15:10', '2024-03-08', 4100.75),  
(4, 'Anil', '2024-03-09', '2024-03-09 14:05:55', '2024-03-15', 1800.00),  
(5, 'Suresh', '2024-01-07', '2024-01-07 23:55:00', '2024-01-12', 2900.00);
```

### 3. CURRENT DATE & TIME FUNCTIONS

```
SELECT CURDATE(); -- current date
```

```
SELECT CURRENT_DATE();
```

```
SELECT CURTIME(); -- current time
```

```
SELECT CURRENT_TIME();
```

```
SELECT NOW(); -- current date + time
```

```
SELECT CURRENT_TIMESTAMP;
```

#### 4. DATE & TIMESTAMP DATA TYPES

Type Stores

DATE YYYY-MM-DD

TIME HH:MM:SS

DATETIME Date + time (no timezone)

TIMESTAMP Date + time (timezone aware)

#### 5. EXTRACTING YEAR, MONTH, DAY

SELECT

YEAR(order\_date),

MONTH(order\_date),

DAY(order\_date)

FROM orders;

#### 6. EXTRACT USING EXTRACT()

SELECT

EXTRACT(YEAR FROM order\_date),

EXTRACT(MONTH FROM order\_date),

EXTRACT(DAY FROM order\_date)

FROM orders;

#### 7. MONTH NAME AND DAY NAME

SELECT

MONTHNAME(order\_date),

DAYNAME(order\_date)

FROM orders;

Example Output

- January, February, March
- Monday, Tuesday, Saturday

## 8. WEEKDAY AND DAY OF WEEK

SELECT

WEEKDAY(order\_date),

DAYOFWEEK(order\_date)

FROM orders;

Difference

Function Range Meaning

WEEKDAY() 0–6 Monday = 0

DAYOFWEEK() 1–7 Sunday = 1

## 9. IDENTIFY WEEKENDS (SATURDAY & SUNDAY)

Method 1: Using DAYNAME

SELECT order\_id, order\_date

FROM orders

WHERE DAYNAME(order\_date) IN ('Saturday', 'Sunday');

Method 2: Using DAYOFWEEK

SELECT order\_id, order\_date

FROM orders

WHERE DAYOFWEEK(order\_date) IN (1, 7);

## 10. IDENTIFY WEEKDAYS

SELECT order\_id, order\_date

FROM orders

WHERE DAYOFWEEK(order\_date) BETWEEN 2 AND 6;

## 11. DATE ARITHMETIC

Add / Subtract Days

```
SELECT order_date, DATE_ADD(order_date, INTERVAL 5 DAY)
```

```
FROM orders;
```

```
SELECT order_date, DATE_SUB(order_date, INTERVAL 3 DAY)
```

```
FROM orders;
```

Add / Subtract Months

```
SELECT DATE_ADD(order_date, INTERVAL 1 MONTH) FROM orders;
```

```
SELECT DATE_SUB(order_date, INTERVAL 2 MONTH) FROM orders;
```

Add / Subtract Years

```
SELECT DATE_ADD(order_date, INTERVAL 1 YEAR) FROM orders;
```

## 12. DATEDIFF (Difference in Days)

```
SELECT
```

```
order_id,
```

```
DATEDIFF(delivery_date, order_date) AS delivery_days
```

```
FROM orders;
```

## 13. TIMESTAMPDIFF (Difference in Units)

```
SELECT
```

```
TIMESTAMPDIFF(DAY, order_date, delivery_date) AS days_diff,
```

```
TIMESTAMPDIFF(MONTH, order_date, delivery_date) AS months_diff
```

```
FROM orders;
```

## 14. LAST DAY OF MONTH

```
SELECT LAST_DAY(order_date) FROM orders;
```

## 15. FIRST DAY OF MONTH

```
SELECT DATE_SUB(order_date, INTERVAL DAY(order_date)-1 DAY)
FROM orders;
```

## 16. DATE FORMAT (IMPORTANT)

```
SELECT DATE_FORMAT(order_date, '%d-%m-%Y') FROM orders;
SELECT DATE_FORMAT(order_date, '%M %d, %Y') FROM orders;
```

### Common Format Specifiers

#### Specifier Meaning

%Y Year (2024)

%y Year (24)

%m Month number

%M Month name

%d Day

%W Weekday name

%H Hour

%i Minutes

%s Seconds

## 17. STRING TO DATE (STR\_TO\_DATE)

```
SELECT STR_TO_DATE('15-01-2024', '%d-%m-%Y');
```

## 18. TIMESTAMP FORMATTING

```
SELECT DATE_FORMAT(order_timestamp, '%d-%m-%Y %H:%i:%s')
FROM orders;
```

## 19. FILTER BY MONTH (JAN, FEB, MAR)

```
SELECT *
```

```
FROM orders
```

WHERE MONTH(order\_date) = 1; -- January

SELECT \*

FROM orders

WHERE MONTHNAME(order\_date) = 'February';

## 20. FINANCIAL YEAR LOGIC (CASE)

SELECT order\_date,

CASE

WHEN MONTH(order\_date) >= 4 THEN CONCAT(YEAR(order\_date), '-',  
YEAR(order\_date)+1)

ELSE CONCAT(YEAR(order\_date)-1, '-', YEAR(order\_date))

END AS financial\_year

FROM orders;

## 21. REAL-TIME BUSINESS USE CASES

Orders in Last 7 Days

SELECT \*

FROM orders

WHERE order\_date >= CURDATE() - INTERVAL 7 DAY;

Orders Placed Today

SELECT \*

FROM orders

WHERE DATE(order\_timestamp) = CURDATE();